Hotel Booking System - HTML Code

Single-file HTML, CSS and JavaScript. Save as hotel_booking.html and open in any browser. Features: room search, availability check, booking form with validation, and booking cancellation.

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1">
<title>Hotel Booking System</title>
<style>
  * { box-sizing: border-box; margin: 0; padding: 0; }
  body { font-family: Arial, sans-serif; background: #f4f6f9; color: #222; }
  header { background: #1f3a5f; color: #fff; padding: 16px; text-align: center; }
  main { max-width: 900px; margin: 20px auto; padding: 0 12px; }
  section { background: #fff; border-radius: 8px; padding: 16px;
            margin-bottom: 16px; box-shadow: 0 1px 4px rgba(0,0,0,.1); }
  h2 { margin-bottom: 12px; font-size: 18px; }
  .row { display: flex; flex-wrap: wrap; gap: 10px; }
  .row > div { flex: 1 1 140px; }
  label { display: block; font-size: 13px; margin-bottom: 4px; }
  input, select { width: 100%; padding: 8px; border: 1px solid #bbb;
                  border-radius: 4px; }
  button { background: #2b7de9; color: #fff; border: 0; padding: 9px 16px;
           border-radius: 4px; cursor: pointer; margin-top: 10px; }
  button:hover { background: #1a63c4; }
  button.danger { background: #d9534f; }
  .room { display: flex; justify-content: space-between; align-items: center;
          border: 1px solid #ddd; border-radius: 6px; padding: 12px;
          margin-bottom: 10px; flex-wrap: wrap; gap: 8px; }
  .price { font-weight: bold; color: #1f7a3f; }
  .msg { padding: 10px; border-radius: 4px; margin-top: 10px; }
  .error { background: #fdecea; color: #a12622; }
  .ok { background: #e6f4ea; color: #1e6b34; }
  table { width: 100%; border-collapse: collapse; font-size: 14px; }
  th, td { padding: 8px; border-bottom: 1px solid #eee; text-align: left; }
  .hidden { display: none; }
</style>
</head>
<body>
<header><h1>Grand Palace Hotel</h1></header>

<main>
  <!-- 1. SEARCH -->
  <section>
    <h2>Search Rooms</h2>
    <div class="row">
      <div><label>Check-in</label><input type="date" id="checkin"></div>
      <div><label>Check-out</label><input type="date" id="checkout"></div>
      <div><label>Guests</label>
        <select id="guests">
          <option>1</option><option selected>2</option>
          <option>3</option><option>4</option>
        </select>
      </div>
    </div>
    <button onclick="searchRooms()">Search</button>
    <div id="searchMsg"></div>
  </section>

  <!-- 2. RESULTS -->
  <section id="resultsBox" class="hidden">
    <h2>Available Rooms</h2>
    <div id="rooms"></div>
  </section>

  <!-- 3. BOOKING FORM -->
  <section id="bookBox" class="hidden">
    <h2>Guest Details</h2>
    <p id="summary" style="margin-bottom:10px"></p>
    <div class="row">
      <div><label>Full name</label><input id="name"></div>
      <div><label>Email</label><input id="email" type="email"></div>
      <div><label>Phone</label><input id="phone"></div>
    </div>
```

```
    <button onclick="confirmBooking()">Confirm Booking</button>
    <div id="bookMsg"></div>
  </section>

  <!-- 4. BOOKINGS LIST -->
  <section>
    <h2>My Bookings</h2>
    <table>
      <thead>
        <tr><th>ID</th><th>Guest</th><th>Room</th><th>Dates</th>
            <th>Total</th><th></th></tr>
      </thead>
      <tbody id="bookings"></tbody>
    </table>
    <p id="noBookings">No bookings yet.</p>
  </section>
</main>

<script>
  // ---------- DATA ----------
  const ROOMS = [
    { id: 1, name: "Standard Room", type: "Standard", price: 80,  cap: 2 },
    { id: 2, name: "Deluxe Room",   type: "Deluxe",   price: 120, cap: 3 },
    { id: 3, name: "Family Suite",  type: "Suite",    price: 180, cap: 4 },
    { id: 4, name: "Executive Suite", type: "Suite",  price: 250, cap: 4 },
    { id: 5, name: "Single Room",   type: "Standard", price: 60,  cap: 1 }
  ];

  let bookings = [];     // kept in memory (use a backend/DB in production)
  let selectedRoom = null;
  let search = null;
  let nextId = 1001;

  // ---------- HELPERS ----------
  const $ = id => document.getElementById(id);

  function nightsBetween(a, b) {
    return Math.round((new Date(b) - new Date(a)) / 86400000);
  }

  function overlaps(aIn, aOut, bIn, bOut) {
    return new Date(aIn) < new Date(bOut) && new Date(bIn) < new Date(aOut);
  }

  function isAvailable(roomId, cin, cout) {
    return !bookings.some(b => b.roomId === roomId &&
                               overlaps(cin, cout, b.checkin, b.checkout));
  }

  // ---------- SEARCH ----------
  function searchRooms() {
    const cin = $("checkin").value;
    const cout = $("checkout").value;
    const guests = parseInt($("guests").value, 10);
    const today = new Date().toISOString().split("T")[0];

    if (!cin || !cout) return showMsg("searchMsg", "Select both dates.", true);
    if (cin < today)   return showMsg("searchMsg", "Check-in is in the past.", true);
    if (cout <= cin)   return showMsg("searchMsg", "Check-out must be after check-in.",
      true);

    search = { cin, cout, guests, nights: nightsBetween(cin, cout) };
    $("searchMsg").innerHTML = "";

    const list = ROOMS.filter(r => r.cap >= guests &&
                                   isAvailable(r.id, cin, cout));
    renderRooms(list);
  }

  function renderRooms(list) {
    $("resultsBox").classList.remove("hidden");
    $("bookBox").classList.add("hidden");
    if (!list.length) {
      $("rooms").innerHTML = "<p>No rooms available for those dates.</p>";
      return;
    }
    $("rooms").innerHTML = list.map(r => `
      <div class="room">
```

```
        <div>
          <strong>${r.name}</strong><br>
          <small>${r.type} | Up to ${r.cap} guests</small>
        </div>
        <div class="price">$${r.price} / night<br>
          <small>Total: $${r.price * search.nights}</small>
        </div>
        <button onclick="selectRoom(${r.id})">Book</button>
      </div>`).join("");
  }

  // ---------- BOOKING ----------
  function selectRoom(id) {
    selectedRoom = ROOMS.find(r => r.id === id);
    const total = selectedRoom.price * search.nights;
    $("summary").textContent =
      `${selectedRoom.name}: ${search.cin} to ${search.cout} ` +
      `(${search.nights} nights) - Total $${total}`;
    $("bookBox").classList.remove("hidden");
    $("bookBox").scrollIntoView({ behavior: "smooth" });
  }

  function confirmBooking() {
    const name = $("name").value.trim();
    const email = $("email").value.trim();
    const phone = $("phone").value.trim();

    if (!name || !email || !phone)
      return showMsg("bookMsg", "Fill in all guest details.", true);
    if (!/^[^@\s]+@[^@\s]+\.[^@\s]+$/.test(email))
      return showMsg("bookMsg", "Enter a valid email.", true);
    if (!isAvailable(selectedRoom.id, search.cin, search.cout))
      return showMsg("bookMsg", "Room was just booked. Search again.", true);

    const booking = {
      id: nextId++,
      roomId: selectedRoom.id,
      roomName: selectedRoom.name,
      guest: name, email, phone,
      checkin: search.cin,
      checkout: search.cout,
      total: selectedRoom.price * search.nights
    };
    bookings.push(booking);

    showMsg("bookMsg", `Booking confirmed! Your ID is #${booking.id}.`, false);
    renderBookings();
    $("name").value = $("email").value = $("phone").value = "";
    $("resultsBox").classList.add("hidden");
  }

  function cancelBooking(id) {
    if (!confirm("Cancel booking #" + id + "?")) return;
    bookings = bookings.filter(b => b.id !== id);
    renderBookings();
  }

  function renderBookings() {
    $("noBookings").classList.toggle("hidden", bookings.length > 0);
    $("bookings").innerHTML = bookings.map(b => `
      <tr>
        <td>#${b.id}</td>
        <td>${escapeHtml(b.guest)}</td>
        <td>${b.roomName}</td>
        <td>${b.checkin} to ${b.checkout}</td>
        <td>$${b.total}</td>
        <td><button class="danger"
             onclick="cancelBooking(${b.id})">Cancel</button></td>
      </tr>`).join("");
  }

  function escapeHtml(s) {
    return s.replace(/[&<>"']/g, c =>
      ({ "&": "&amp;", "<": "&lt;", ">": "&gt;",
         '"': "&quot;", "'": "&#39;" }[c]));
  }

  function showMsg(id, text, isError) {
```

```
    $(id).innerHTML =
      `<div class="msg ${isError ? "error" : "ok"}">${text}</div>`;
  }

  // ---------- INIT ----------
  (function init() {
    const today = new Date().toISOString().split("T")[0];
    $("checkin").min = today;
    $("checkout").min = today;
  })();
</script>
</body>
</html>
```